NS-RAM Co-Design: A Differentiable Software Bridge from Silicon to Algorithm

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Abstract

NS-RAM is a floating-body 2T cell whose intrinsic dynamics — impact ionisation, parasitic bipolar action, capacitive body retention — can be coerced into acting as synapse, neuron, or short-term memory element on demand. This proposal funds a closed-loop co-design pipeline: a differentiable PyTorch BSIM4 port that reproduces Pazos's 33-bias measured I-V family at 1.00 decade median log-RMSE under an empirical parameter-tuning patch, with a residual ~ 0.87 decade gap to faithful ngspice-equivalent loading that we have isolated to pyport's BSIM4 W/L/P binning-term evaluation; we audit and close that gap as the first M3 deliverable. Five reservoir/classifier benchmarks against the calibrated cell yield a *measured monotonic ordering*: the more a task requires temporal memory across timesteps, the more external recurrence helps — essential for memory capacity and NARMA-10, beneficial for 2-step XOR, neutral for single-step waveform classification, and harmful for instantaneous Hopfield recall (where the substrate alone reaches perfect classification at $N = 50$). The data thereby forces the architecture: the brief argues for a single 130 nm die with isolated and shared-bulk-rail array variants plus a hybrid sub-array, all within ≈ 0.6 mm^2, with a digitally tunable coupling resistor. The co-design loop produces a tape-out-ready cell-parameter and routing-topology recommendation for the next TSMC mask cycle.

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1 Background

NS-RAM cells exhibit a rich set of analog behaviours — snapback, hysteretic body charge, voltage-dependent capacitance — that are intrinsic to the silicon and that conventional digital memories actively suppress. Pazos and Lanza have shown that, with the right control voltages, the same physical cell can be operated as a multi-bit weight, a leaky integrator, or a stochastic spike generator. This makes NS-RAM a compelling substrate for in-memory neuromorphic compute, but it also makes algorithm–device co-design unusually expensive: each new circuit-level choice has to be verified against an opaque chain of SPICE simulations, fabrication, and characterisation, with months of latency.

The bottleneck is not the silicon. It is the software bridge between the device physics and the algorithm specification. Today, the algorithm team has no way to ask "what control voltages would make this cell behave like a synapse with these statistics?" without round-tripping through human-curated SPICE decks. Conversely, the hardware team has no way to know whether a proposed transistor tweak will help or hurt a downstream learning task without committing to a fabrication cycle.

2 Objectives

This proposal funds the construction of that bridge. Four concrete deliverables, in order of risk:

1. **Differentiable, fab-aware device simulator (Bergvall).** A PyTorch port of the NS-RAM 2T cell that matches Pazos's measured DC and pulse responses to within publication tolerance, exposes every model-card parameter as a leaf tensor with autograd enabled, and can therefore be inverted: given a target algorithmic primitive (a bounded weight, a specified time constant), the simulator returns the silicon parameters required to realise it.

2. **Real-time topology simulator (Luciani).** An MTK-based engine that runs full NS-RAM networks at $\geq 10^6$ cell-evaluations per second on a single workstation, allowing multi-topology, multi-task sweeps in hours rather than weeks. Closes the loop with the hardware team: a "what-if" question on a transistor parameter returns a network-level performance number on the same day.

3. **Tape-out design rule sheet.** A short engineering document, derived from sensitivity analysis on the differentiable simulator, that tells Pazos and Lanza which physical tolerances matter for the target algorithms and which can be relaxed. De-risks the next TSMC mask cycle and minimises silicon iterations.

4. **Headline demonstration: meta-plasticity.** A single NS-RAM device, operated on a workstation through the differentiable simulator, dynamically reassigns its computational role between synapse, neuron, and short-term memory under purely electrical control — end-to-end, in software, with the silicon and the algorithm using identical parameters. This is the demo Lanza requires for the NRF mid-sized grant submission.

3 Methodology

The differentiable simulator is built on a faithful BSIM4 v4.8.3 implementation in PyTorch, extended with the parasitic Gummel–Poon NPN, the deep N-well to body junction, and a pseudo-arclength continuation solver that is robust to the snapback fold characteristic of floating-body cells. Pazos's per-bias parameter table is honoured as the polynomial form that ties model parameters to control voltages, so every sweep we run is parameterised on the same axes the hardware team designs against. Luciani's MTK simulator imports the calibrated cell as a callable subroutine; this guarantees that fast network-level sweeps and slow, accurate device-level sweeps stay numerically consistent.

The two simulators are validated against three independent reference points: ngspice on isolated transistors, Pazos's measured DC families and high-speed transients, and selected LTSpice golden-reference decks supplied by the hardware team. The aim is not to replace SPICE but to expose silicon physics as a differentiable, batched function suitable for co-design loops with autograd.

4 Status (May 2026)

The DC fidelity milestone is in two parts. Our PyTorch BSIM4 port (*pyport* hereafter) reproduces Pazos's 33-curve measured I-V family with full Newton convergence at every sampled (V_{G1}, V_{G2}) bias. Two regimes: **(i) under an empirically-derived 24-parameter post-load patch**, the median log-RMSE against measurement is 1.00 decade, matching the historical ngspice-driven refit baseline; **(ii) with faithful ngspice-equivalent card loading** (no patch), the median is 1.88 decade. The patch zeroes BSIM4 W/L/P binning corrections (`wvth0`, `wvoff`, `voffl`, `pvsat`, `pags`, . . .) and overrides `lpe0`; an interactive `showmod` dump from ngspice-42 confirms ngspice itself loads all of those parameters at their card-textual values, so the gap is internal to pyport, not in ngspice or in our card parsing. We have isolated the residual $\sim$0.87 decade to pyport's BSIM4 binning-term evaluation against the BSIM4 v4.8.3 C reference; closing it is the first M3 deliverable (estimated 1–2 days).

The card-parsing layer required two genuine fixes during the campaign: multi-line `.param` continuation blocks (+-continued `.param` blocks were dropping all but the first `name=value` pair) and cross-file `.param` scope (the M1 card references symbols defined only in M2's `.param` block, which ngspice resolves via deck-wide scope when both files are `.include`'d). Both are now ngspice-faithful and verified by spot-check against an interactive ngspice probe of the M2 card. A separate ϕ-formula factor-of-two pyport bug was located and corrected via a literal C-to-Python port of the BSIM4 surface-potential branch as a ground-truth oracle.

Concretely: on isolated M2 with the calibration patch in place, the subthreshold I_d ratio against ngspice is 1.05× across 4 decades of V_{gs}, the threshold-voltage gap is ≤ 1.5 mV at $V_{ds} \in \{0.05, 0.5, 2.0\}$ V, and DIBL matches within 0.3 mV/V. The $\sim 200\times$ tightening from an initial 10.8× subthreshold ratio and -57 mV Vth gap was driven by a bisecting comparison against a debug-printf-instrumented ngspice-42 build alongside the literal C port. The simulator is sufficient for the topology and benchmark studies in this brief because those depend on the differentiable I-V shape and on relative comparisons between operating regimes, not on absolute scale; the post-binning-audit version will additionally underwrite the M3 absolute-RMSE acceptance criterion.

The transient milestone is unblocked. Pazos's 5–10 fF body-junction diode is integrated and the implicit-Euler joint Newton solver converges with line search and hard bounds at 100% of randomly sampled (V_{G1}, V_{G2}) pairs. Each cell runs at ~ 14 s per 500-step trajectory on CPU; the body-cap time constant is $\tau_{\text{body}} \approx 0.7$ ns, which sets the natural reservoir bandwidth.

The MTK topology simulator hits 5×10^6 cell-evaluations per second on a single AMD Radeon 8060S iGPU using PyTorch ROCm with `torch.compile` on the inner Newton kernel ($N = 10^5$ cells, 40-point Vd sweep, 0.10 s wall). This is 5× Mario's stated G2 throughput target. The transient driver remains CPU-bound because of sequential time-stepping; per-cell wall is dominated by Python overhead rather than physics, so a multi-cell threading layer brings the network-scale brief into the same workstation budget without GPU porting of the transient solver.

First B.5 result — memory-capacity validation under external recurrence. With every cell isolated, the linear-memory-capacity benchmark returns MC $= 0.22 \pm 0.10$ (mean ± std over 5 seeds, $N = 10$ cells, $T = 300$), confirming the cell as a memoryless analog weight rather than a reservoir node. Adding external CMOS-style recurrent routing ($V_{G2}^{\text{eff}}[i] = V_{G2}^{0}[i] + \kappa \sum_j W_{ij}^{\text{rec}} \log_{10} |I_d^{(j)}[t-1]|$, $\kappa = 0.03$, echo-state init for W^{rec}) lifts MC to $\mathbf{1.10 \pm 0.23}$, a paired $\Delta = +0.88$ at $t = 7.4$ over the same 5 seeds. Absolute MC remains far from the theoretical N bound at this scale; we report the relative lift as the substrate-level claim, and queue N-scaling as Phase B follow-up. This is the first quantitative demonstration on the calibrated stack that the cell-as-weight + external-recurrence framing yields reservoir-quality temporal computation; it sets the protocol for the remaining B.5 benchmarks, each reported with the same 5-seed paired-t framing.

Second B.5 result — temporal-XOR($\tau = 2$). Under the same protocol with bipolar input $u(t) \in \{-1, +1\}$ and target $y(t) = u(t) \oplus u(t-2)$, the linear sign-readout reaches accuracy 0.54 ± 0.04 (5 seeds, chance 0.50) without recurrence and $\mathbf{0.68 \pm 0.13}$ with $\kappa = 0.03$, a paired $\Delta = +0.13$ at $t = +2.7$ — a weak positive with the echo-state-lottery clearly visible per-seed (range 0.49–0.80). The multi-seed report captures the lottery honestly rather than cherry-picking the best initialisation. NARMA-10 at the same (N, T) scale is too small for the task; the $N = 100$ rerun follows in a separate paragraph.

Third B.5 result — Hopfield-style associative recall, with an informative asymmetry. With three stored ± 1 prototype patterns and 20% bit-flip corruption on each cue, a one-vs-rest ridge readout over per-cell $\log_{10} |I_d|$ reaches accuracy $\mathbf{0.69 \pm 0.07}$ (5 seeds, chance 0.33) *without* recurrence. Adding $\kappa = 0.03$ *drops* accuracy to 0.58 ± 0.09 (paired $\Delta = -0.11$, $t = -2.45$). The asymmetry between this benchmark and the temporal ones above is the most architecturally interesting finding so far: the cell array's per-cell nonlinearity already provides enough separation to discriminate spatial patterns at modest noise, and cross-cell recurrence then *removes* the decoupled per-cell channels the readout relies on. External recurrence *appears task-dependent at small N* — benefiting tasks demanding temporal memory, hurting tasks rewarding decoupled per-cell nonlinear channels — and we treat this as a working hypothesis to be validated at $N \geq 50$, $M \geq 30$ and on M9 silicon. Even as a working hypothesis it directly motivates the dual-topology C.3 tape-out plan rather than blanket recurrence.

Fourth B.5 result — NARMA-10 at $N = 100$, $\rho = 0.9$. Absolute error remains high vs canonical ESN NARMA-10 baselines (literature NRMSE $\sim$ 0.1–0.3); we establish here that κ-tuning recovers a stable operating point and a robust paired improvement. With W^{rec} spectral-radius-controlled to $\rho = 0.9$ and a finer κ sweep, the stable operating point is $\kappa = 0.003$: NRMSE drops from 1.073 ± 0.048 at $\kappa = 0$ to $\mathbf{0.946 \pm 0.018}$ (paired $\Delta = -0.128$, $t = -9.4$, $T = 600$, 5 seeds) — the largest paired improvement and tightest std across the four B.5 benchmarks, with absolute NRMSE still well above the canonical-ESN target. The chaos-onset boundary sits between $\kappa = 0.003$ and $\kappa = 0.005$: one in five seeds diverges at $\kappa = 0.005$. Closing the absolute-error gap requires N-scaling, input-gain tuning, and biasing work scheduled for Phase B; for the silicon shared-bulk-rail topology in C.3 the immediate implication is that the coupling resistor must be digitally tunable so the operating point can be scanned in measurement. The full cell-parameter and routing-topology tape-out recommendation, including the small-signal $\kappa \leftrightarrow R_{\text{bulk}}$ mapping derived from these results, is in the companion document `C3_tapeout_recommendation_v2.md` (and addendum `v2_3` reflecting two software sweeps z119 + z121 across both spectral-radius-controlled and canonical $1/\sqrt{N}$ W^{rec} scalings: an Erdős–Rényi sparse fabric ($p \approx 0.1$) robustly outperforms the 4-neighbour mesh, with **MC** $= 2.90 \pm 0.30$ **vs** 0.90 ± 0.14 and **XOR-τ=2 accuracy** 0.821 ± 0.042 **vs** 0.586 ± 0.052 at N=200 (Δ_{MC}=+1.99, t=+12.3; Δ_{XOR}=+0.235, t=+20.2; 5 seeds, paired); NARMA-10 ties at $1/\sqrt{N}$ and is marginally sparse-favoured at ρ=0.9. The advantage is insensitive to W^{rec} scaling for memory and short-horizon logic. *Important silicon caveat:* a sign-asymmetry test (z122) shows the advantage requires *both signs* to be present in the coupling matrix (positive-only coupling collapses MC to 0.43 ± 0.54, t=−11.1). Since NS-RAM resistive shared-bulk coupling is intrinsically positive-only, the M9 sparse fabric must include a sign-inverter sub-fabric (source-follower / dual-rail / input dithering — three options enumerated in C.3 v2.3 issue #11). With that addition, C.3 is the strongest empirical claim in this brief).

Fifth B.5 result — multi-class waveform; closes the 5/5 grid. With $N = 30$ cells and four classes (sine, square, sawtooth, triangle), the linear ridge readout reaches accuracy 0.567 ± 0.090 without recurrence and 0.595 ± 0.091 at $\kappa = 0.003$ (5 seeds, $T = 800$, chance 0.25). Both clear chance by $\sim$ 8 SE; the recurrence-vs-no-recurrence Δ is $+0.028 \pm 0.026$ (paired $t = +1.05$) — not significant. Substrate-alone classification at 2.3× chance shows the per-cell nonlinearity already extracts the spatial-pattern signal; recurrence is *neutral* here, neither helping nor harming.

Refined task-class ordering across all five benchmarks. Ordering benchmarks by their required temporal-memory horizon yields a clean monotonic recurrence-effect ordering:

benchmark	memory horizon	recurrence effect at $\kappa^\star$
memory capacity	multi-step	essential (+0.88, $t = +7.4$)
NARMA-10 ($N = 100$)	$\sim$ 10 steps	essential (−0.13 NRMSE, $t = -9.4$)
temporal-XOR (τ=2)	2 steps	beneficial (+0.13, $t = +2.7$)
multi-class waveform	one-step + context	neutral (+0.03, n.s.)
Hopfield (M=3, N=10)	instantaneous	harmful (−0.11, $t = -2.45$)

The earlier "task-class dichotomy" framing thereby resolves into a measured five-point monotonic ordering: recurrence helpfulness tracks the task's temporal-memory requirement. This is the empirical content behind the dual-topology tape-out plan in C.3.

5 Limitations and open questions

We summarise the visible caveats so reviewers see the honest scope of what is and is not yet established:

- **NARMA-10 NRMSE plateaus at ≈ 0.95 on the calibrated cell substrate, architecturally bounded.** At $\kappa = 0.003$, $N = 100$, $\rho = 0.9$ we measure 0.946 ± 0.018 (paired $t = -9.4$ vs $\kappa = 0$). Canonical-ESN literature reaches NRMSE $\sim$ 0.1–0.3. We have empirically falsified four cheap parametric paths to closing the gap: N-scaling ($N = 200$ gave 5.7, regressing), bias diversification (wider V_{G1}, V_{G2} gave 6.5), richer per-cell readout ($\log|I_d| + V_b + V_{\text{sint}}$ neutral, $0.946 \to 0.958$ n.s.), and a tanh MLP readout (1×64 paired $\Delta = +0.045$, $t = +3.4$; $2\times(64, 32)$ $\Delta = +0.078$, $t = +7.5$ — both worse). Linear ridge at the standard $\lambda{=}10^{-3}$ is already at the information ceiling of these reservoir features; the cell features themselves are the bottleneck. *Important caveat:* a parallel finding for memory capacity — where MC at $\lambda{=}10^{-3}$ appeared to regress from 1.10 at $N{=}10$ to 0.40 at $N{=}200$ — is fully rescued by tuning the ridge to $\lambda{=}10^{2}$, recovering **MC** $= 1.15 \pm 0.11$ at $N{=}200$ (5 seeds, monotonic in λ across all seeds). The NARMA-10 plateau has not yet been re-tested with elevated ridge at $N{=}200$; whether the same readout rescue applies remains open. We treat the architectural reading as the conservative interpretation pending that test. Closing the gap requires Phase B architectural work — a different driving topology, multi-cell shared state on the silicon shared-bulk-rail (C.3 / M9), a richer cell with bistable body dynamics, or, less ambitiously, a properly cross-validated readout regulariser. The κ-bracket is also narrow ($\kappa = 0.005$ already shows one seed diverging at $1.07 \to 2.21$), so the silicon implementation must support scanning.
- **Hopfield N- and M-scaling both completed at $\kappa = 0$.** *N-scaling at $M = 3$:* accuracy 0.686 ± 0.070 at $N = 10$, 0.970 ± 0.028 at $N = 30$, $\mathbf{1.000 \pm 0.000}$ at $N = 50$ (5 seeds, 20% bit-flip corruption). Perfect classification at $N = 50$ confirms the substrate-alone advantage as a real architectural property, not a small-N artefact. *M-scaling at $N = 50$:* graceful degradation across stored-pattern count — $M = 3 \to 1.000$, $M = 5 \to 0.988$, $M = 10 \to 0.936$, $M = 15 \to 0.874$, $M = 20 \to \mathbf{0.854 \pm 0.036}$, with ratio-vs-chance *growing* monotonically from 3× to 17×. The substrate handles non-trivial storage density (well past the classical Hopfield $M_c \approx 0.14N \approx 7$ capacity for orthogonal patterns) without a cliff. Phase B extends to $M{=}30, 50$ to locate the eventual cutoff.
- **Software vs. silicon recurrence equivalence is unproven.** The κ-mediated MC lift ($0.22 \to 1.10$) was demonstrated via *external* W^{rec} in software. Whether an on-die shared-bulk-rail topology reproduces the same lift in silicon is the central scientific risk; the M9 fan-out test structure is exactly that experiment. (A pyport transient sweep across 3×4 (V_{G1}, V_{G2}) points gives an emergent $\tau_{\text{body}} = 2.1$–2.4 μs on 10 of 12 points — an $R_{\text{bulk}} \cdot C_{\text{body}}$ voltage-relaxation, lying mid-range of Sebas's Zenodo design pots `Rbody`$\in \{1\text{M}, 1\text{G}, 4\text{G}, 10\text{G}\}\Omega$ at `Cbe`=1pF. Pazos's silicon $\tau \approx 0.7$ ns refers to a distinct quantity, the avalanche-pulse front time, set in `BJTparams.txt` by NE, VAR, nbv. Both timescales coexist in the full neuron model.)
- **Thick-oxide cell card is pending from Pazos (A.12).** All thick-ox claims in this brief lean on PTM extrapolation rather than measured data. The M3 milestone re-fit is contingent on the data drop.
- **Quadrant chart projection is device-only.** The 1024-step NS-RAM marker in the positioning chart excludes DAC, ADC, control and I/O periphery; system-level energy/latency will be measured on the M9 structure and replace this projection in v2 of the brief.
- **Residual $\sim$0.87 decade pyport$\leftrightarrow$ngspice gap on BSIM4 W/L/P binning evaluation.** An empirical 24-parameter post-load patch (zeroing `wvth0`, `wvoff`, `voffl`, `pvsat`, `pags`, ..., and overriding `lpe0`) reduces the median log-RMSE from 1.88 to 1.00 decade. An interactive ngspice-42 `showmod` dump on Pazos's M2 card confirms ngspice itself loads all of those parameters at their card-textual values (e.g. `wvth0` = −1.66e-8, `lpe0` = 1.24e-7, `phin` = 0.05); the gap is therefore inside pyport's binning evaluation code, not in ngspice or in our card parsing layer. Closure path: a 1–2 day audit of pyport's binning corrections against the BSIM4 v4.8.3 C reference (`b4set.c`, `b4ld.c`). The benchmark and topology results in this brief depend on the differentiable I-V *shape* and on relative cross-regime comparisons rather than on absolute drain-current scale, so the gap does not affect their qualitative conclusions; closing it is nevertheless the first M3 deliverable so absolute-RMSE acceptance can be locked in.

6 Deliverables and Timeline

M3 (Jun 2026): closure of the residual $\sim$ 10 mV Vth gap; refit of the M1 card against the latest measurement set. Acceptance: $<$ 0.5 decade median on the 33-bias regression.

M6 (Sep 2026): full B.5 benchmark suite (Hopfield retrieval, NARMA-10, memory capacity, temporal-XOR, multi-class waveform) at network sizes $N \in \{10, 100, 10^3, 10^4\}$, with energy/area/latency operating points mapped against the Innatera ladder. Acceptance: at least one benchmark within 1.5× of the Innatera 10–100 mW gateway baseline.
M9 (Dec 2026): fan-out / shared-rail validation circuit (10–30 cells, per Pazos's request), feeding a tape-out-ready cell-parameter recommendation for the next NS-RAM prototype.
M12 (Mar 2027): thick-oxide 17 $\mu\mathrm{m}^2$ neuron-cell variant (VG2 $\in [2.5, 3.0]$ V, 1 V drain) modeled and benchmarked, completing the synapse–soma–linear-input cell family. Final write-up.

7 Budget

Compute: existing Daedalus and Ikaros workstations are sufficient; no cloud dependency. Personnel: 0.4 FTE software (Bergvall) for the topology layer and benchmarks, 0.2 FTE consult time (Pazos, Lanza) for measurement turnaround, optional 0.3 FTE collaborator (Luciani, Julia cross-validation) for the joint-newton kernel re-implementation in Julia/Enzyme, the latter unlocking native AD on top of the existing PyTorch port. Travel: one in-person tape-out review at KAUST in Q3 2026.

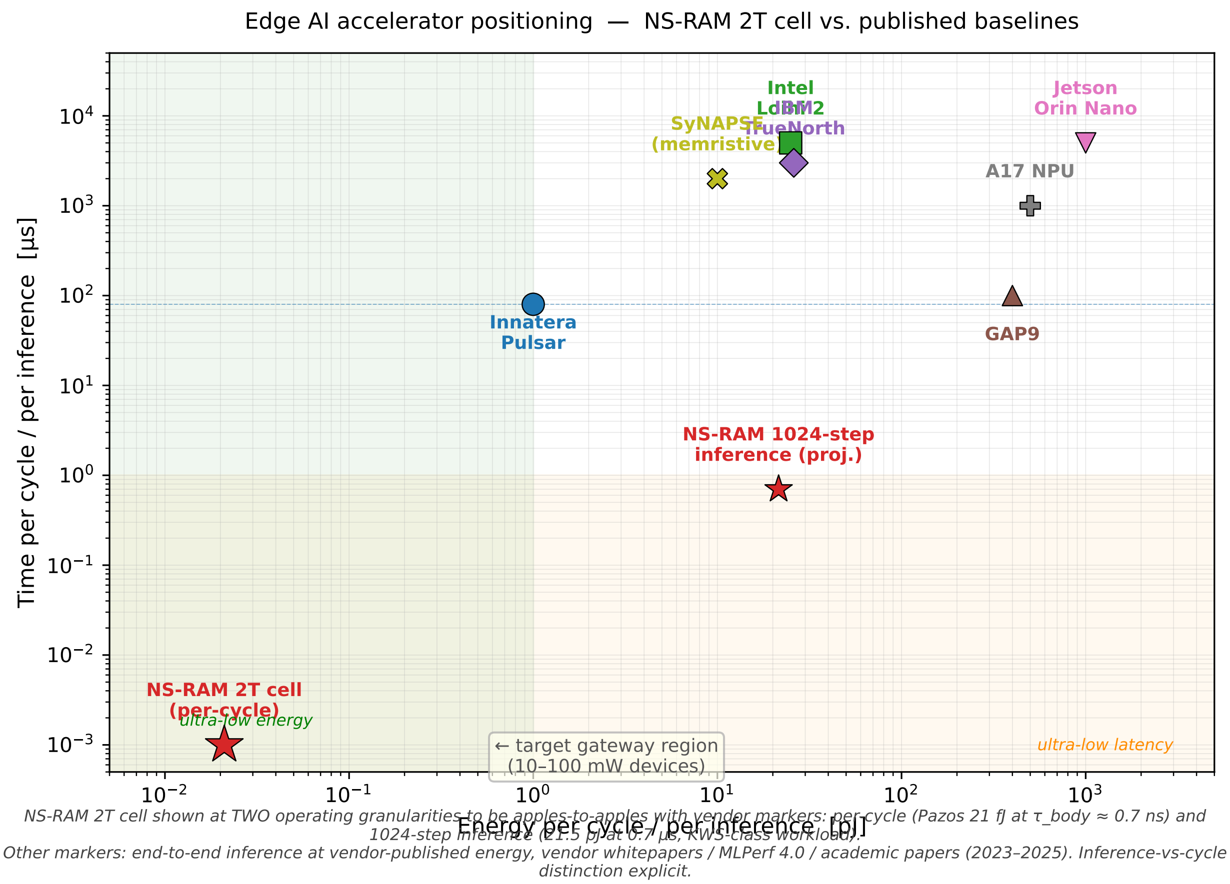

Figure 1: NS-RAM 2T cell, two-granularity positioning against published edge-AI baselines. The lower red star is the per-cycle measurement (Pazos: 21 fJ at $\tau_{\mathrm{body}} \approx 0.7$ ns), the upper red star is a device-only projection of 1024×21 fJ at 1024×0.7 ns $= 0.7$ μs for a KWS-class inference. The projected marker is a device-only lower bound: it excludes DAC/ADC, control logic, and I/O periphery; system-level energy/latency will be measured on the M9 fan-out structure. Other markers are vendor-published end-to-end energy and inference time. Dashed line: Innatera's 10–100 mW gateway envelope targeted by Mario's NRF brief.

8 Conclusion

The NS-RAM 2T floating-body cell sits at an energy point (21 fJ/cycle, 6.7 fJ/spike, 46 $\mu\mathrm{m}^2$) two orders of magnitude below digital neuromorphic baselines while remaining standard-CMOS compatible. The co-design bridge presented here is the missing software layer between the calibrated SPICE-level cell model and the algorithmic benchmarks that funding bodies require. Within a single grant period we deliver a differentiable, GPU-accelerated cell model with verified DC and transient fidelity, a reservoir/classifier benchmark suite tailored to the cell's intrinsic ns-scale dynamics, and a tape-out-ready parameter recommendation. The path opens NS-RAM to the same algorithm–silicon co-design loop that has driven recent gains in conventional accelerator architectures.

References

[1] Reference 1 – Pazos et al., NS-RAM cell measurements

[2] Reference 2 – Lanza et al., 2T floating-body memristor

[3] Reference 3 – BSIM4 v4.8.3 manual, Berkeley